

## HOW TO PERFORM YOUR ROLE

### Intended Learning Outcome

The objectives of CUH601CMD Software Engineering coursework are derived from the module objectives. Following the completion of the project, students are expected to be able to:

1. Evaluate an architectural design against accepted quality criteria.
2. Select, evaluate and use tools and techniques to successfully manage a large-scale software project, including configuration management and version control.
3. Use a range of appropriate tools to contribute to the development of a solution to a real-world problem.
4. Select, evaluate and apply standards, tools and techniques for assuring software quality.
5. Identify and engage with relevant social, professional, legal and ethical issues on a practical computing project.

### The Project

The assessment for CUH601CMD Software Engineering is team-based. However, each team member will be assigned a relevant portion of individual tasks that will contribute to the completion of the team's project. Each team consists of 4 to 6 students, who will be working on the same project. The team must select one of the two study cases below.

**Komodo Hub** – A Digital Platform for Community-Supported Animal Conservation [Refer the module AULA page]

**JustInTime** – FloatFry's Manufacturing Resource Planning [Refer the module AULA page]

This module emphasizes a comprehensive software engineering approach to development. Therefore, the construction/coding aspect of the development of the proposed solution needs to be supported by equally strong evidence of practical project management and process control. To meet the objectives, students are asked to contribute to the creation of their team's comprehensive project report (~5000 words, citation excluded) covering three main aspects: **Process Control, Product Development, and Project Management**. Details on how to form and manage your team, how to manage your project including scheduling tasks and activities, the structure of the report document and what should be included in its content are available in the **HOW TO** documents on the module AULA page.

## Your Role and Activities

There are three types of role you can play in your team: a **Product Owner**, a **Scrum Master**, and **Developers** (either System Architect, Database Engineer, Web Developer, or Cloud Engineer). You may work independently whilst performing a role. However, your activities **MUST** contribute to your team's work for the completion of the project. The table below lists the activities you can perform in your role, either weekly, as needed, or as a one-off.

| EVENTS                                                                                                                                                      |                        | ROLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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|                                                                                                                                                             |                        | PRODUCT OWNER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SCRUM MASTER                                                                                                                                                                                                                                                                                                          | DEVELOPERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>PROCESS CONTROL:</b> all things related to how your team perform SCRUM appropriately, including Scrum meetings, Sprint Review, Sprint Retrospective, etc |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                                                                                                             | <b>SPRINT PLANNING</b> | <ul style="list-style-type: none"> <li>Generating product goal(s) – <i>start of Project</i></li> <li>Communicating product goal(s) – <i>as needed</i></li> <li>Developing Product Backlog – <i>start of Project</i></li> <li>Communicating Product Backlog – <i>as needed</i></li> <li>Refining Product Backlog if there is a change – <i>in between Sprints</i></li> <li>Mapping Product Backlog items to product goal(s) – <i>start of Project and when there are changes</i></li> <li>Assigning values to the items – <i>start of Project and when there are changes</i></li> <li>Identifying and determining new items to be included in the Sprint – <i>start of Sprint</i></li> </ul> | <ul style="list-style-type: none"> <li>Facilitating Product Owner to communicate product goal(s) – <i>as needed</i></li> <li>Facilitating Product Owner to communicate Product Backlog – <i>as needed</i></li> <li>Double checking and updating the status of items in the Product Backlog - <i>weekly</i></li> </ul> | <ul style="list-style-type: none"> <li>Generating Sprint goal(s) – <i>start of Sprint</i></li> <li>Communicating Sprint goal(s) – <i>as needed</i></li> <li>Developing Sprint Backlog from Product Backlog – <i>start of Sprint</i></li> <li>Communicating Sprint Backlog, i.e., giving arguments on how items fit the goals and can increase values and utility – <i>as needed</i></li> <li>Decomposing Sprint Backlog items into jobs - <i>start of Sprint and weekly</i></li> <li>Generating/updating work plan to complete the jobs – <i>start of Sprint and weekly</i></li> </ul> |

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|                                                                                                                                                                                    | <b>WEEKLY SCRUM</b>         | <ul style="list-style-type: none"> <li>• Checking the status of each item - <i>weekly</i></li> <li>• Acknowledging/disregarding weekly increments - <i>weekly</i></li> </ul>                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Facilitating developers to develop Sprint Backlog - <i>weekly</i></li> <li>• Checking the status of the Sprint - <i>weekly</i></li> </ul>                                               | <ul style="list-style-type: none"> <li>• Checking the status of each item in the Sprint Backlog - <i>weekly</i></li> <li>• Reporting the status of the current Sprint run – <i>weekly</i></li> <li>• Presenting weekly increments before the stakeholders - <i>weekly</i></li> <li>• Adjusting Sprint Backlog – <i>as needed</i></li> </ul> |
|                                                                                                                                                                                    | <b>SPRINT REVIEW</b>        | <ul style="list-style-type: none"> <li>• Inspecting the outcome of the Sprint – <i>end of Sprint</i></li> <li>• Acknowledging test report – <i>end of Sprint</i></li> <li>• Adjusting Product Backlog – <i>as needed</i></li> <li>• Identifying new items to include in the next Sprint Backlog – <i>end of Sprint</i></li> </ul> | <ul style="list-style-type: none"> <li>• Facilitating developers to present increments before the stakeholders – <i>end of Sprint</i></li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>• Presenting Sprint increments before the stakeholders – <i>end of Sprint</i></li> <li>• Presenting test report – <i>end of Sprint</i></li> </ul>                                                                                                                                                    |
|                                                                                                                                                                                    | <b>SPRINT RETROSPECTIVE</b> | <ul style="list-style-type: none"> <li>• Concluding the current Sprint – <i>end of Sprint</i></li> </ul>                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• Evaluating the current Sprint covering individual team members, obstacles, team communication, artifacts produced, effectiveness – <i>every day, report at end of Sprint</i></li> </ul> | <ul style="list-style-type: none"> <li>• Giving valuable feedback about the current Sprint – <i>end of Sprint</i></li> </ul>                                                                                                                                                                                                                |
| <b>PRODUCT DEVELOPMENT:</b> all things related to how your team design and develop the product, including analysing requirements, designing the software, modeling the design, etc |                             |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                             |

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|  | <p><b>REQUIREMENT ANALYSIS</b></p> | <ul style="list-style-type: none"> <li>Analysing case – <i>start of Project and as needed</i></li> <li>Identifying stakeholders – <i>start of Project</i></li> <li>Identifying requirements – <i>start of project</i></li> <li>Documenting requirements specifications – <i>start of Project and when there are changes</i></li> </ul> | <ul style="list-style-type: none"> <li>Analysing case – <i>start of Project and as needed</i></li> <li>Identifying stakeholders – <i>start of Project</i></li> <li>Identifying requirements – <i>start of project</i></li> </ul> | <ul style="list-style-type: none"> <li>Analysing case – <i>start of Project and as needed</i></li> <li>Identifying stakeholders – <i>start of Project</i></li> <li>Identifying requirements – <i>start of project</i></li> </ul> |
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|  | DESIGN |  |  | <ul style="list-style-type: none"> <li>Analysing business processes requirements – <i>start of Project and as needed</i></li> <li>Modeling processes: Use Case Diagram – <i>start of Project and as needed</i></li> <li>Modeling processes: Activity Diagram – <i>start of Project and as needed</i></li> <li>Analysing data structure requirements – <i>start of Project and as needed</i></li> <li>Modeling data: DFD – <i>start of Project and as needed</i></li> <li>Modeling data: ERD/Class Diagram – <i>start of Project and as needed</i></li> <li>Analysing system architecture requirements – <i>start of Project and as needed</i></li> <li>Modeling software architecture: Class Diagram – <i>start of Project and as needed</i></li> <li>Modeling hardware architecture: Component Diagram – <i>start of Project and as needed</i></li> <li>Analysing system security requirements – <i>start of Project and as needed</i></li> <li>Determining security measures – <i>start of Project and as needed</i></li> </ul> |
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|                                                                                                                                                                                     | DEVELOPMENT        |                                                                                                                                                                                                                                                           |                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Developing interface prototype – <i>as needed</i></li> <li>Developing database prototype – <i>as needed</i></li> </ul>                                                                                                                                                                                                              |
|                                                                                                                                                                                     | TESTING            | <ul style="list-style-type: none"> <li>Developing verification points – <i>weekly and as needed</i></li> <li>Conforming acceptance test - <i>weekly</i></li> </ul>                                                                                        |                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Identifying test scenarios from verification points - <i>weekly</i></li> <li>Designing acceptance test cases covering usefulness and ease of use - <i>weekly</i></li> <li>Developing test data - <i>weekly</i></li> <li>Modeling test environment - <i>weekly</i></li> <li>Generating test report – <i>end of Sprint</i></li> </ul> |
| <b>PROJECT MANAGEMENT:</b> all things related to how your team manage the project, including setting up milestones, scheduling tasks and activities, assigning jobs to members, etc |                    |                                                                                                                                                                                                                                                           |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                     | PROJECT DEFINITION | <ul style="list-style-type: none"> <li>Developing project vision – <i>start of Project</i></li> <li>Developing project mission/objective – <i>start of Project</i></li> <li>Determining project scope and boundaries – <i>start of Project</i></li> </ul> | <ul style="list-style-type: none"> <li>Identifying risks – <i>start of Project</i></li> <li>Managing work tools – <i>start of Project and as needed</i></li> </ul> |                                                                                                                                                                                                                                                                                                                                                                            |

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|  | <b>SCHEDULING</b>           | <ul style="list-style-type: none"> <li>Identifying milestones – <i>start of Project</i></li> </ul> | <ul style="list-style-type: none"> <li>Identifying tasks – <i>start of Project</i></li> <li>Decomposing tasks into jobs – <i>start of Project</i></li> <li>Valuing jobs in unit-days – <i>start of Project</i></li> <li>Identifying resources – <i>start of Project</i></li> <li>Assigning resources to jobs - <i>weekly</i></li> <li>Managing milestones – <i>as needed</i></li> <li>Creating project schedule – <i>start of Project, update as needed</i></li> </ul> |                                                                                         |
|  | <b>MEETING ORGANIZATION</b> | <ul style="list-style-type: none"> <li>Attending meetings – <i>as needed</i></li> </ul>            | <ul style="list-style-type: none"> <li>Creating meetings – <i>Scrum Meeting weekly, Sprint Review end of Sprint, Sprint Retrospective end of Sprint, others as needed</i></li> <li>Facilitating all members to attend and make contributions to meetings – <i>as needed</i></li> <li>Attending meetings – <i>as needed</i></li> </ul>                                                                                                                                  | <ul style="list-style-type: none"> <li>Attending meetings – <i>as needed</i></li> </ul> |
|  | <b>MINUTES OF MEETINGS</b>  |                                                                                                    | <ul style="list-style-type: none"> <li>Generating minutes of meetings – <i>as needed</i></li> <li>Compiling and organizing documentation - <i>weekly</i></li> <li>Completing and finishing up project final report – <i>end of Project</i></li> </ul>                                                                                                                                                                                                                  |                                                                                         |